Pedal Ratio Documentation

For our next iteration of our FS car to pass the brake test and achieve simultaneous lockup of all the 4 wheels, several factors were considered and solved to get the most optimal value of pedal ratio and brake bias for our car.

This was done and optimised while keeping several factors in the mind:

- All the 4 wheels should lock up almost simultaneously to pass the brake test and ensure proper braking of the car.
- There should be a sufficient time before the wheels lock and the car completely stops (visible to the naked eye) so that the judges can clearly see that indeed all the 4 wheels are locked first then the car comes to a complete stop.
- We want to minimize the pedal ratio (get a lower value) as much as possible while adhering to the above mentioned conditions to minimize the weight and "bulkiness" of the brake pedal.

To solve this problem, several differential equations were solved to get to a convergent solution while adhering to the condition of achieving simultaneous lockup of all 4 wheels.

Initial Parameters

```
R = 0.2034;                                  % Tire Radius (m)
m = 311.07;                                  % Total mass (Kg)
mf_unsprung = 16.2;                          % Front Unsprung mass (Kg)
mr_unsprung = 27.6;                          % Rear Unsprung mass (Kg)
ms = m - (mf_unsprung + mr_unsprung);        % Total sprung mass (Kg)
g = 9.81;                                    % gravitational acc (m/sec^2)
rwb = 0.55;
wb = 1.5495;                   % wheelbase (m)
a_cg = wb*rwb;                 % CG to front axle (m)
b_cg = wb*(1-rwb);             % CG to rear axle (m)
ClA = 4.84;
CdA = 1.80;
front_aero_bias = 0.62;
rho = 1.225;                   % density (Kg/m^3)
h_cg = 0.3;                    % CG height (m)
pitch_center = 0.106;          % Pitch centre height (m)
kf = 45000;                    % Front spring stiffness (N/m)
kr = 55000;                    % Rear spring stiffness (N/m)
kt = 90000;                    % tire stiffness (N/m)
mr_ratio = 1.1;                % Motion ratio
kwf = ((kf/(mr_ratio^2))*kt)/(kt + (kf/(mr_ratio^2))); % front wheel rate
kwr = ((kr/(mr_ratio^2))*kt)/(kt + (kr/(mr_ratio^2))); % rear wheel rate
x01 = ms*g*(1-rwb)/(2*kwf);          % Initial compression at front wheels
x02 = (ms*g*rwb)/(2*kwr);            % Initial compression at rear wheels
cf = 2500;                     % Front Damping coefficient (N-sec/m)
cr = 2500;                     % Rear Damping coefficient (N-sec/m)
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cwf = cf/(mr_ratio^2);              % front damping at wheel
cwr = cr/(mr_ratio^2);              % rear damping at wheel
pad_disc_fc = 0.45;
area_mc = 154e-6;                   % cross section area of master cylinder (m^2)
area_calliper_front = 794e-6;       % cross section area of front brake pad (m^2)
area_calliper_rear = 510e-6;        % cross section area of rear brake pad (m^2)
brake_force_driver = 600;           % driver force on brake pedal (N)
R_disc_front = 0.076;               % front brake disc radius (m)
R_disc_rear = 0.073;                % rear brake disc radius (m)
I_wheel = 0.266;                    % Inertia of wheel (Kg-m^2)
Iy = 200;                           % pitch inertia (Kg-m^2)
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The torque provided by the callipers to the front and rear discs can be formulated as:
Tb_f = *2(brake_driver_force/area_mc)(area_calliper_front)(pad_disc_fc)(R_disc_front)(r_opt)(1-b_opt)*
Tb_r = *2(brake_driver_force/area_mc)(area_calliper_rear)(pad_disc_fc)(R_disc_rear)(r_opt)(b_opt)*

Where “r_opt’ is the optimal pedal ratio and “b_opt” is the optimal rear brake bias to be estimated.

Mathematical Equations

1. *ax_calculated = (2*Fx_f_final_val + 2*Fx_r_final_val - F_drag_end)/m;*
   *where F_drag_end is equal to 0.5*rho*ClA*v_end^2*

2. *dv_dt = ax;*

3. *dw_f_dt = (-Tb_f - Fx_f*R)/I_wheel;*
   *dw_r_dt = (-Tb_r - Fx_r*R)/I_wheel;*

4. *M_longitudinal = 2*(-Fx_f - Fx_r)*(h_cg - pitch_center); % -Fx is positive (forward force on chassis)*
   M_suspension = -a_cg(2*kwf*(a_cg*pitch_angle + x01) + 2*cwf*a_cg*pitch_rate) ...*
   + b_cg(2*kwr*(x02 - b_cg*pitch_angle) - 2*cwr*b_cg*pitch_rate);*

5. *d_pitch_rate_dt = (M_longitudinal + M_suspension) / Iy;*

6. *d_pitch_dt = pitch_rate;*

Cost Function

Now, all the above equations are to be solved simultaneously such that the following cost function is minimized to ensure the lockup of all the 4 wheels simultaneously and minimize the overall speed of the vehicle at the time of lockup of all 4 wheels but big enough to visually observe that "all 4 wheels are locked simultaneously”.

```
% OBJECTIVE 1: Minimize Vehicle Speed at Lockup (Max Deceleration)
velocity_penalty = v_end^2;
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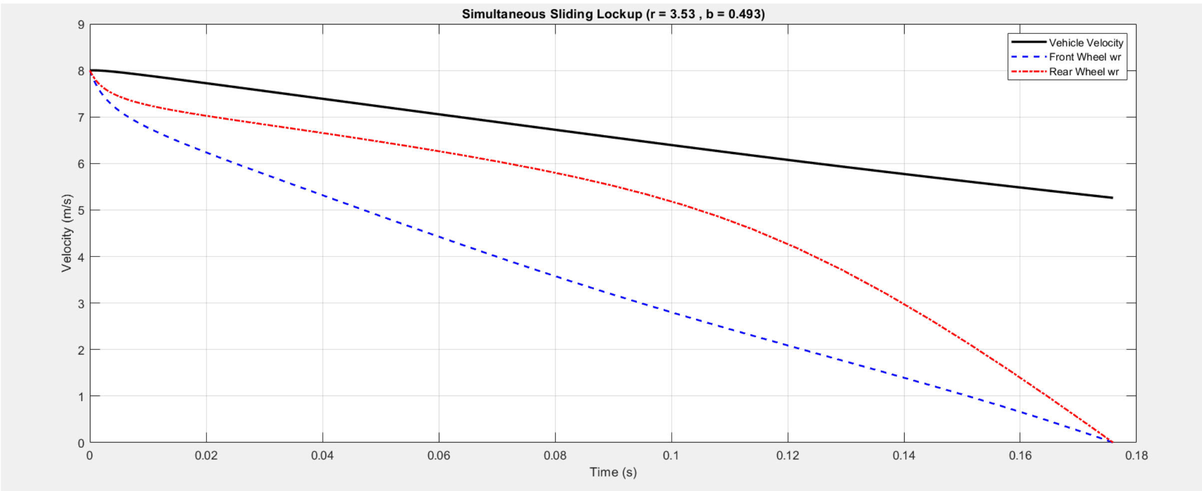
% OBJECTIVE 2: Simultaneous Lockup
bias_penalty = 10(wf_end^2 + wr_end^2) + 10000*(wf_end - wr_end)^2;*

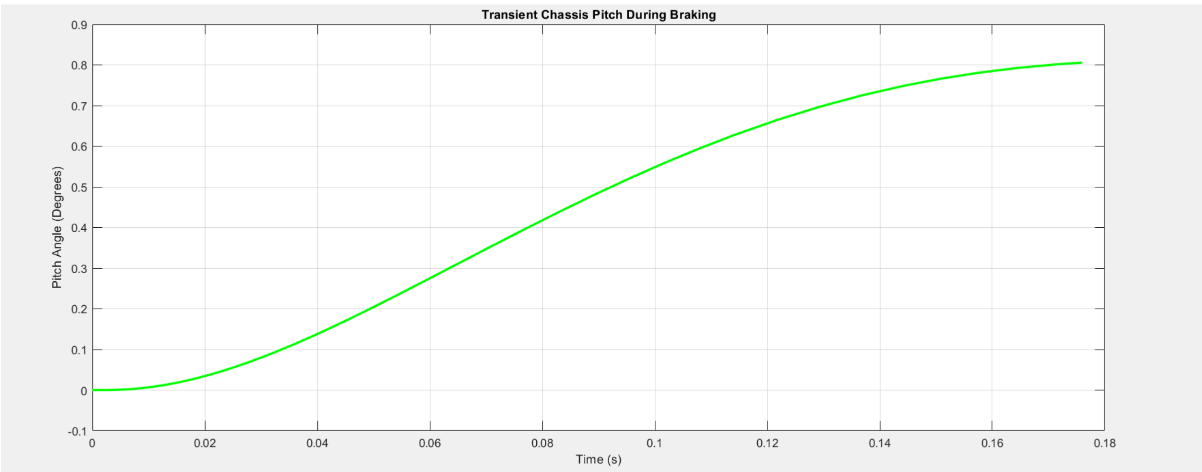
% Add the delay penalty to the final cost
*cost = 2*velocity_penalty + bias_penalty*

Results

For our D2 car with 16” tires and a velocity of 8 m/sec at the start of brake test, we obtain:

- Pedal Ratio = 3.53
- Rear Weight Bias = 49.3%
- Front Brake Torque = 379.25N
- Rear Brake Torque = 227.09N
- Average Deceleration = -1.59g





Full Code:
https://docs.google.com/document/d/1MHBSQuQ7IxJPPB7GeaGXxaS_HyCTO5zRuK9idqifuDY/edit?usp=sharing